BANIK BUDDHADEB

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Academic Background

Degree	Field of Study	Institution	Country	Date
B.E.	Electronics & Communication Engineering	Dayananda Sagar College of Engineering	India	2025–2029

CGPA: 8.9 / 10

Description of Activities

Projects

SIH 2026 – Anti-Drone System (repo)

- Contributing to a multi-subsystem anti-drone platform for Smart India Hackathon 2026, combining radar and camera-based detection, sensor data fusion, and STM32-based control for real-time drone tracking.
- Implemented the vision-based detection pipeline using YOLOv8-Nano (~6.2 MB), chosen for its low compute footprint to run inference reliably on less powerful edge hardware.
- Working across the target-tracking pipeline that drives a 2-axis gimbal from fused radar and vision data through to target lock and system response.

FocusDesk – Deadline & Focus Clock (repo)

- Built a desk-based deadline and Pomodoro timer on ESP32 with dual I2C OLED displays, task list, environment monitoring (BME280), and a self-hosted local web dashboard for task sync, needing no app or cloud.
- Refactored a 1300-line monolithic UI class into a per-screen `App` interface architecture, cutting the core navigation file to ~250 lines and making new screens a drop-in addition.

ESP32 FIFA Live Scoreboard (repo)

- Built a real-time football scoreboard on ESP32 that polls the football-data.org API over Wi-Fi and renders live scores on an SSD1306 OLED in a broadcast-style layout, with automatic score/goal updates.
- Integrated a BME280 sensor for environmental readouts alongside score data, with page rotation between scoreboard and environment views on a single display.

ESP32-CAM Timelapse Generator (repo)

- Built a timelapse camera system on ESP32-CAM that captures JPEG frames at fixed intervals to an SD card, with automated filename generation and frame-buffer management for continuous long-run capture.
- Documented a full build-and-usage guide covering hardware setup, capture workflow, and frame-to-video conversion.

Achievements and Other Activities

- 1st Place, E-Summit – conducted by the Entrepreneurship Cell, in association with the DERBI Foundation, for developing a product and its marketing.
- Member, Point Blank – Embedded Systems Club, Dayananda Sagar College of Engineering.

Areas of Expertise

Embedded systems, Computer Vision, Firmware, PCB design, IoT, C/C++.

Technical Skills

Languages: C, C++, Python
Embedded Platforms: Arduino, ESP32, ESP32-CAM, Arduino Nano ESP32, STM32
Protocols & Interfaces: I2C, UART (Serial Communication), USB
Tools: KiCad (PCB Design), Git, GitHub, Linux, Multisim, MATLAB, Oscilloscope
Concepts: Embedded Systems Design, Real-Time Firmware, IoT, Sensor Integration, Computer Vision (YOLOv8-Nano)